

From Research Question to Data: How Arguments, Hypotheses, Measurements, and Data Connect

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1 The Architecture of Quantitative Research

Imagine you are planning to build a house. Most of us would not start by ordering bricks and lumber. We tend to begin with a vision of what we want, then draw blueprints, then work out what materials the plan calls for — and only then start building. Research tends to work in much the same way. You need some sense of what you want to know (the research question), a blueprint (the argument), the construction details (hypotheses and measurements), and then, eventually, the raw materials (the data).

Nonetheless, students often want to skip ahead to the data, and the impulse is an understandable one: it is exciting to see how things actually relate and whether you were right in your prediction. But a questionnaire written before you have thought about what your variables are, how to measure them, and why they might be related will rarely deliver what you were hoping for. This is not only a novice problem. Experienced researchers can make a version of the same mistake. A Research team might, for instance, collect household-level data when the question really calls for individual-level data. That situation is difficult to repair after the fact, since you cannot disaggregate from low resolution into high-resolution data.

Figure 1 sketches the sequence of steps this chapter outlines. Your research question helps settle the choice of dependent variable. Your argument shapes which hypotheses are sensible. Argument and hypotheses together inform how you measure your concepts, which in turn points to the unit of analysis, the population you sample from, and so on.

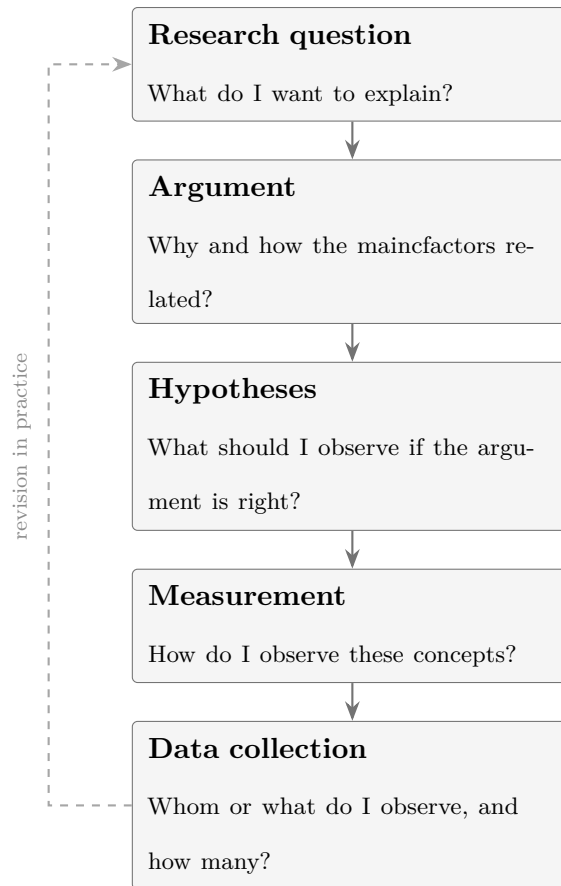


Figure 1: The architecture of quantitative research. Each stage tends to shape the one below it, though some degree of revision upward is common in practice.

1.1 Description, Explanation, and Prediction

Before expanding on the steps identified above, it is worth briefly noting that not every study you encounter will follow this sequence, since studies aim to do different things. It helps to separate three basic categories of quantitative research.

1.1.1 Description

Description tells us what the world looks like. How many households fall below the poverty line, how support for a party has shifted over a decade, how combatants in a conflict differ from non-combatants in age and education. No causal claim is being made; we are characterizing a state of affairs.

Description is also where a good deal of discovery starts. In June 1981 a short bulletin reported five cases of an unusual pneumonia among young men in Los Angeles. It offered no mechanism and no

theory, only a cluster of rare cases. The work that followed identified HIV as the underlying cause. Tycho Brahe spent decades recording planetary positions more precisely than anyone before him but never quite settled on a theory of the heavens; it was Kepler, working off of Brahe's tables, who found the laws. The famous John Snow's map of cholera deaths in Soho did not offer anything about the underlying cause, but it made the pattern clear, which in turn revealed the potential source of the disease and laid groundwork for developing theory of waterborne transmission of disease. And in 1985 Farman and his colleagues described a hole in the ozone layer that theory had not predicted and that instruments had been discarding as measurement error. In each case the description set the groundwork for the explanation.

But there are limits to what can reasonably be claimed on the basis of descriptive data, and in practice those limits present a real temptation to stretch findings well past what they can support. Description typically tells you a snippet of what happened rather than why.

Consider data showing that wildfires have become more frequent and more destructive in some regions over recent decades. Taken as simple description, this is a solid finding. But the graph on its own says little about the cause, and it is tempting — especially on issues we genuinely care about — to attach a larger claim about why the pattern appears and present that alongside the data as though it were part of the finding. We might conclude, for instance, that the trend is broadly due to climate change. That may well be so, but until it is properly tested it remains a hypothesis, because the same pattern is consistent with other causal accounts: a) A century of fire suppression allowed fuel to build up in forests that had historically burned more often. b) Housing has expanded into fire-prone land, raising both the number of ignitions and the damage any given fire does. c) Detection has improved as well, so satellite coverage now records fires that earlier decades missed. Several of these may be operating together, and sorting out their relative contributions is what explanatory studies are for and it simply cannot be gauged directly from the trend line.

1.1.2 Explanation

Explanation asks why. In the example we expand on later in this chapter, Humphreys and Weinstein ask why some individuals join rebel forces in civil conflicts while others do not. This is why arguments become indispensable, since an explanation is a claim about the reasoned causal chain behind it, and that chain is rarely visible in the data directly.

Explanation as a research goal also contributes more to the systematic advancement of knowledge

than description alone. A description is limited to the population and period visible to us, so a new decade or a different country calls for a new set of data because we cannot simply extrapolate from the original data without knowing how and why.

Take the democratic peace research. The descriptive observation is that democracies have rarely gone to war with one another. On its own that is a striking regularity. Once scholars proposed mechanisms for the empirical regularity, the finding started generating more expectations. If the reason is institutional constraint — leaders who must answer to electorates and legislatures face higher costs for losing wars — then the effect should weaken where those constraints are low, and we should see something similar in the behavior of leaders facing other kinds of accountability. If instead the reason is a shared norm of resolving disputes through bargaining, then the pattern should hold between democracies even when institutional constraints are weak, and newly democratized states should behave differently from established ones. Those are different theory driven predictions about different cases, and neither would be possible to gauge from the correlation alone.

In another case, suppose a study finds that access to microfinance raises household income in rural Bangladesh. As a description, that is a fact about rural Bangladesh during the years studied. But if the reason it works is that loans let households pay for investments they otherwise could not afford, then the effect should be largest where borrowing is hardest, absent where people can already borrow cheaply from family or local moneylenders, and smaller for households with nothing to invest in. Each of those is a testable claim about a case nobody has looked at yet. The explanation turned one finding into a much broader research program.

Findings accumulate in this way. Each new study can extend a mechanism into a new domain, refine it and clarify its limits. The failed cases are often the most informative.

The contact hypothesis is a useful illustration. The starting observation was puzzling in that contact between members of different groups sometimes reduced prejudice and sometimes worsened. A Harvard personality psychologist, Gordon W. Allport specified in 1954 the conditions under which contact should be beneficial for group dynamics and prejudice: roughly equal status between the groups in the situation, shared goals, cooperation rather than competition in pursuit of them, and support from institutions or authorities. The claim was no longer that contact reduces prejudice. It was that contact reduces prejudice under identifiable circumstances. Allport held the conditions to be more or less necessary, warning that contact falling short of them would mean “the more contact the more trouble.” However, half a century of accumulated studies suggested instead that

they strengthen the effect without being essential. The disagreement remains open and the research program continues to develop over time.

The strength of a theoretical argument and the limitations of description also shows clearly when the data alone are misleading. Two examples are worth keeping in mind. The first is the classic problem of self-selection. Participants in a job training programme often earn less afterwards than non-participants, and taken at face value the comparison suggests that training does harm. A theory of who is likely to enrol specifies why the finding makes sense: people sign up precisely because their prospects are poor, so the two groups should be compared carefully.

The second is that arguments tell you where to look. In the rebellion example, we expand on later, the grievance framework and the selective incentives framework predict different things about who joins — the most deprived under one account, those with moderate opportunities under the other. Without those competing arguments in hand, a researcher confronted with a survey of ex-combatants has no principled reason to examine one variable rather than another, and any pattern that emerges can be narrated after the fact. The theory is what turns an analysis into a test.

This is why the sequence in Figure 1 is presented here as the standard case of scientific research, and why the chapters that follow are organized around it. The framing is not meant to dismiss the alternatives. As we laid out, descriptive work identifies the phenomena worth explaining in the first place, and predictive work can be enormously useful where a decision has to be made and the mechanism is somebody else's problem. But both are comparatively limited in what they let us build on. A well-supported explanation is something other researchers can take up, test elsewhere, and either strengthen or break, and that activity is most of what we call scientific progress.

1.1.3 Prediction

Prediction, or forecasting, asks what is likely to happen next, or what is likely to hold in a case we have not yet seen. Which districts stand at elevated risk of unrest next year? Which patients are likely to relapse? Of the three purposes, this is the one whose standing as science continues to be hotly discussed.

The complaint is not that forecasts are unimportant. It is that what makes a forecasting model good has often little to do with understanding. A variable is included in the model to improve performance on cases the model has not seen, rather than why it needs to be included. In each

instance the question is whether the forecast is accurate and timely enough to guide a decision.

The applications are nonetheless consequential and everywhere. Credit scoring assigns risk of default from repayment histories, balances, and the age of an account. Clinical scores estimate the chance of a cardiac event within the decade, and hospital warning systems flag patients drifting toward sepsis hours before a clinician might notice. Humanitarian agencies rely on forecasts of political violence by country and district to plan their deployments. Firms anticipate which customers are about to leave, epidemiologists project case counts, and election forecasters turn polls into probabilities.

The most instructive contemporary case is the large language model (llm), which carries this logic to its limit. Its training objective is prediction in the strictest sense: given a stretch of text, assign specific probabilities to whatever might come next. Nothing in that objective asks why one continuation should be likelier than another. There are no selected variables at all. The features are learned from an enormous corpus of text and information rather than specified, and distributed across an enormous number of parameters. A model of this kind absorbs regularities in text without regard to what produces what, so treating it as associations.

Yet the same example also reveals the complex nature of how understanding is formed. A view now seriously held is that prediction, pursued at great enough scale, may resemble understanding. To anticipate text about the physical world reliably, the argument runs, such system may represent how that world behaves and arrive at explanatory structure as a by-product. Those who study the internal functioning of llm models report finding organized representations rather than mere surface associations, but whether such representations amount to understanding is vigorously contested. Critics object that fluent imitation of explanatory language is not explanation, and that a system may reproduce the form of an account while grasping nothing. The dispute is unresolved, and certainly won't be solved here. The lesson for the us a researchers is narrower: the line between predicting and explaining, though real and worth maintaining, is not altogether as clean as some may like, and thus we should approach these debates with a sense of open mind and curiosity.

To illustrate the lesson more specifically, forecasting has also been used creatively to serve distinctly theoretical ends. Ward, Greenhill, and Bakke examined the leading accounts of how civil wars begin and found, for example, that variables with large and confidently estimated coefficients contributed almost nothing to the models' capacity to anticipate which countries would in fact descend into conflict. In another case, well over a hundred research teams were given abundant longitudinal early childhood records on thousands of children and asked to predict such later outcomes as

academic attainment and material hardship. Even the strongest submissions predicted poorly, and none appreciably surpassed a rudimentary benchmark. What forecasting models revealed in this circumstance is that our understanding of social phenomena appear far less foreseeable and therefore generalizable into the future than the academic literature suggests and that certainly clarifies the limits of our understanding of complex social processes.

1.2 Research is Messy in Practice

Finally, it is worth emphasizing that research in practice is messier than this diagram. Findings prompt new questions, an unexpected pattern may send you back to the theory, and much valuable work is frankly exploratory. Exploratory analysis that generates a hypothesis and explanatory analysis that tests one are both useful, but they carry different evidentiary weight, and issues primarily arise when we confuse the first for the second.

For instance, a relationship you noticed while exploring the data is still worth reporting; it simply gets reported as something to investigate rather than something you have established or tested. It is key to modulate your language: “these results suggest a pattern that warrants further testing” rather than “these results confirm our prediction”.

Further, as you browse through work across various disciplines, you will likely encounter variations of the approach we introduced here. Some subfields expect numbered hypotheses stated before any results appear; others treat that as artificial and prefer an argument developed alongside the evidence. Reading recent articles in the journals you hope to publish in will tell you more about the prevailing norms than any general methodological advice, including this chapter’s.

None of this means that approaches departing from the strict sequence should be rejected outright. Inductive and exploratory work has generated a great deal of what we know, and insisting that every finding arrive as a preregistered confirmation would leave much of social science unwritten. The requirement is narrower and more manageable: be clear about how your findings should be interpreted, and about how much they can bear. A pattern discovered in the data is a genuine contribution when presented as such, and a liability only when dressed up as a test it never was.

2 The Research Question

Reading is key to developing interesting and nuanced questions. This must sound like an obvious statement, the kind you would normally expect to hear from your parents, but in practice reading is more often than not treated as a chore. Interesting questions tend to emerge because you have been participating in a sort of conversation among scholars and thinkers. Reading work — good or bad — trains your sense of what a good argument looks like. This ability to think is absorbed rather than taught, and it cannot be sufficiently supplemented with structured coursework. Naturally, you also learn from these “conversations” which claims are contested, and, most importantly, which of your ideas are actually new rather than merely new to you.

Reading does help in other ways too. You are learning what datasets exist and who maintains them, how long comparable projects took to field, how other researchers measured the complex concepts you also care about, what sample sizes were sufficient to detect the effects you are interested in, and where previous authors ran into trouble.

Finding a manageable research question is the critical first step in any academic research project. In quantitative research, scholars face the additional challenge of needing accessible datasets or planning a potentially time-consuming data collection. The desire to answer big, important questions often conflicts with the reality of limited time, limited resources, and other practical constraints.

However, research questions are not set in stone. In practice, we start with a vaguer idea and as we develop the argument in more detail, we sometimes alter or refine the research question to some degree even at later stages of a project. When I speak to students, they often worry that once they specify their first initial research question that they may not alter it in any other way. But there may be challenges in the process, where the dataset you planned to use is no longer available or authors who previously promised to share their dataset change their minds. However, regardless of the issues that you can encounter as a researcher, it is important to take care developing your question and articulating it well, including developing a real in depth understanding of the topic where you can minimize the changes later in the process.

2.1 The Anchor: Identifying Your Dependent Variable

The dependent variable is your outcome of interest — the thing you are trying to explain. It serves as the anchor for your entire study, and without a clear anchor your research will drift.

Consider the difference between three different types of framing of a research interest:

Important

- A. “I’m interested in social media and teenagers.”
- B. “I want to study the effect of social media on teenagers.”
- C. “I want to explain why some teenagers develop depression.” “How does social media use affect teenage depression?”

The first statement identifies a general topic area but provides no direction. Are you trying to explain social media use? Teenage behavior? The relationship between the two?

The second statement identifies social media as the independent variable and teenagers as the population of interest, but it fails to specify any outcome you are trying to explain. The word “effect” implies causation, but effect on *what*? This vagueness creates a counterproductive abundance of possibilities. Social media could affect depression, anxiety, self-esteem, academic performance, sleep quality, body image, social skills, attention span, political attitudes, friendship quality, risk-taking behavior, cyberbullying experiences, or dozens others. Each of these requires a fundamentally different argument about causal mechanisms.

The third framing identifies depression as the dependent variable — the outcome to be explained. Now you have an anchor. You know you are trying to explain variation in depression. Social media might be only one explanatory factor, but you have a clear target, which immediately focuses your thinking.

2.1.1 How Dependent Variables Shape Arguments

Different dependent variables require different arguments, even within the same topic. Suppose you are interested in education and socioeconomic status. If the outcome is **years of schooling completed**, the argument concerns why some people stay in school longer: family resources, the opportunity cost of a child’s labour, local school availability. If it is **test scores**, you are no longer explaining duration but performance, and the argument shifts to instruction quality, home learning environments, or the stress of economic insecurity. If it is **choice of major**, neither of those arguments applies; now you need gender socialization, expected returns, or exposure to role models.

2.1.2 The Meandering Problem

Without a clear outcome, arguments drift. A student interested in poverty and crime might write that poverty breeds desperation and crime, that crime drives out business and deepens poverty, and that declining neighbourhoods transmit a culture of violence to the young. Each step is plausible. But the dependent variable has changed three times — from individual offending, to neighbourhood decline, to intergenerational transmission — and testing these claims would require different data at different levels of analysis.

Specifying the outcome first keeps the argument coherent: *why do individuals commit property crime?* Neighbourhood and economic factors still matter, but as influences on individual decisions rather than outcomes in their own right. Narrowing further, to adolescents, would change the theoretical story again.

3 The Theoretical Argument

Once you have your research question and your dependent variable, the next step is formulating your argument. An argument is fundamentally an explanation of **why and how** two or more phenomena relate to one another in the population. Unlike a simple assertion — “I think poverty causes crime” — an argument includes the *mechanism* explaining why the phenomena under study may or may not be related. What is it about poverty that would lead to crime? Is it desperation for resources? Is it the breakdown of social institutions? Is it relative deprivation compared to wealthier neighbors? Each of these suggests a different causal mechanism and would lead to different predictions about when and where we would see the relationship.

You have two main approaches to developing an argument. You can articulate an argument that has been proposed in previous studies and that you wish to test with a new set of data or in a new context. This is the more common approach — building directly on existing theory and research. Alternatively, you can present your own original theoretical idea, though even here you will likely be building on previous studies.

3.1 An Illustrative Example: Why People Join Rebellions in Civil Wars

Let us look at a concrete example that shows what a well-developed argument looks like in practice. Humphreys and Weinstein, in their 2008 study of participation in Sierra Leone’s civil war, needed to

explain why some individuals joined rebel groups while others did not. Rather than just asserting that certain factors matter, they laid out three distinct theoretical frameworks drawn from decades of scholarship on conflict and mobilization.

3.1.1 Framework 1: Economic and Political Grievances

Their first theoretical perspective focuses on grievances as drivers of participation. Scholars of social revolution have long argued that the depth of an individual's discontent with their economic position in society, or their exclusion from political power, is a major causal factor differentiating participants in rebellion from nonparticipants. When this discontent aggregates across individuals in a particular social class or ethnic group, it provides the foundation for mobilization and the onset of violence against the state.

This grievance-based argument has many variants, each emphasizing different elements of individual motivation for joining.

Social class. One variant identifies social class as the critical variable differentiating those who rebel from those who remain on the sidelines or choose to defend the status quo. Karl Marx proposed that the industrial proletariat would be the main engine of revolution against capitalist systems, owing to individuals' shared experiences of exploitation. However, the actual locus of participation in revolutions — poor, rural people rather than the urban working class — shifted scholarly debate toward making distinctions among rural populations.

Agrarian class conflict. Jeffrey Paige, in his analysis of agrarian revolutions, concluded that wage-earning peasants drive rebellion in contexts where landlords, dependent on income from the land, are less able or willing to assent to peasant demands. The mechanism here is economic conflict between classes with incompatible interests over control of the agricultural surplus.

Subsistence crisis. James Scott's description of rebellion in Southeast Asia focused on the subsistence crisis among peasants, demonstrating how population growth, capitalism, and the growing fiscal claims of the state pushed rural residents to the edge of survival. Here the mechanism is desperation: when people face genuine threats to subsistence, rebellion becomes preferable to starvation or destitution.

Dispossession. Intensive study of Latin American revolutions suggested that access to land, rather than absolute poverty, serves as the main indicator of class position. Timothy Wickham-Crowley

argued that peasants physically dislocated from land by elites, or those without access to it in the first place — squatters, sharecroppers, and migrant laborers — are the most prone to revolt. The mechanism shifts from absolute deprivation to dispossession: it is not poverty per se, but being pushed off land or denied access, that creates revolutionary grievance.

Relative deprivation. Others have challenged this focus on land, suggesting that income inequality is the prime source of discontent and motivator of participation. This shifts the mechanism from absolute to *relative* deprivation: people rebel not because they are poor in absolute terms, but because they see others with more and perceive this gap as unjust.

Ethnic difference. A related strand focuses on ethnic and political marginalization rather than economic class. For some scholars, the logic of ethnic mobilization begins and ends with long-standing cultural practices that distinguish ethnic groups. Differences between groups, sometimes reflected in a history of animosity, are believed to make conflict more likely. This argument generates the expectation of ethnically homogeneous factions in which identity is the key determinant of participation. The mechanism is identity-based solidarity.

Modernization and group position. For other scholars, the interaction of ethnic difference with the process of modernization creates the conditions for political violence. The upward social mobility made possible in an environment of economic change often rewards some groups over others, ultimately producing ethno-nationalist and separatist sentiments. Here the mechanism is relative group position: not ancient hatreds, but contemporary competition for resources and status that mobilizes ethnic identities for violent purposes.

Anomie and blocked opportunity. A third variant focuses on personal dislocation and the frustrations arising from an individual's inability to express concerns through normal nonviolent channels. Robert Merton emphasized *anomie* as a source of deviant behavior, as individuals use illegitimate means to attain goals such as wealth, power, or prestige that are valued in their societies but unavailable through legitimate channels. The mechanism is blocked opportunity: when conventional paths to success are closed, people turn to unconventional means, including violence.

Political marginalization. More recently, scholars describing conflicts in West Africa have emphasized how the weakening of social structures can account for the rise of violence. Paul Richards, in a cogent critique of simplistic accounts of African violence, pointed to the growing isolation of most citizens from the loci of political decision-making. The mechanism is political

marginalization: when people feel completely shut out of the political processes that affect their lives, violence becomes the only available form of political expression.

3.1.2 Framework 2: Selective Incentives

The second major theoretical perspective focuses on selective incentives — the material benefits and opportunities that rebel organizations offer to potential recruits. This approach draws on rational choice theories and examines how individuals weigh the costs and benefits of participation.

Rather than viewing rebellion as driven primarily by grievances or ideological commitment, this perspective emphasizes how armed groups recruit by offering tangible rewards: opportunities for looting, regular pay or food, protection from violence, social status and power within the organization, or access to resources controlled by the armed group. The mechanism is instrumental calculation — individuals join not necessarily because they support the cause, but because participation offers material advantages they could not obtain otherwise.

This framework helps explain why rebel groups often recruit heavily among unemployed young men, those with limited economic opportunities, or those displaced by conflict who have lost their livelihoods. The opportunity cost of joining is low (they have little to lose economically), while the potential benefits — immediate material gains, adventure, status — can be substantial. The theory predicts that individuals are more likely to join when the selective incentives offered are attractive relative to their outside options.

3.1.3 Framework 3: Social Sanctions and Networks

The third perspective examines social sanctions and network effects: how social pressure, community norms, and existing relationships influence participation decisions. This approach recognizes that individuals do not make decisions in isolation but are embedded in social networks that can facilitate or constrain their choices.

Armed groups often recruit *through* social networks, with existing members bringing in friends, relatives, or community members. The mechanism involves both positive inducements (joining allows you to remain with your social group and provides protection for your community) and negative sanctions (failure to join when your peers do might mark you as a coward, bring shame on your family, or make you vulnerable to retribution).

This perspective also emphasizes how community-level factors shape individual decisions. In some communities, joining a particular armed group becomes normalized or even expected, particularly when that group provides security or services the state cannot. Social sanctions against non-participation — ostracism, accusations of cowardice or disloyalty, threats to family members — can make joining the path of least resistance even for individuals who lack strong grievances and are not attracted by selective incentives. The theory predicts that individuals embedded in networks with existing combatants, those from communities where one armed group is dominant, or those facing social pressure from peers and family are more likely to join, independent of their individual grievances or material calculations.

3.2 What This Example Teaches Us

Look at what Humphreys and Weinstein accomplished in laying out their argument. They presented not one simple story but multiple theoretical perspectives, each with its own logic about why people join rebellions. They showed how these arguments developed through scholarly debate, how they relate to each other, and where they agree and disagree. They identified the specific mechanisms through which various factors — economic and political marginalization, selective incentives, and social sanctions — are theorized to influence the decision to join a rebellion.

This is what a good argument looks like. It does not just assert relationships; it explains them. It does not just cite studies; it extracts and articulates their theoretical logic. It acknowledges complexity and alternative explanations rather than pretending one simple story explains everything.

Summary

When developing your own argument, do not just say “I expect X to affect Y .” Say: “I expect X to affect Y because of this specific process — when X occurs, it triggers this mechanism, which leads to Y through these steps.” The more specific you are about the causal mechanism, the better positioned you will be to derive testable hypotheses and choose appropriate measurements.

4 From Arguments to Hypotheses

Once you have developed your argument — your theoretical explanation for why and how phenomena relate to each other — you need to translate that argument into testable hypotheses. A **hypothesis**

is a specific, testable prediction derived from your theoretical argument. It states what you expect to observe in your data if your argument is correct.

This translation from argument to hypothesis is not optional and cannot be skipped. It is the bridge between abstract theory and concrete empirical analysis. Without clear hypotheses, you do not know what you are testing. With poorly formulated hypotheses, you cannot tell whether your data supports or refutes your argument.

4.1 What Makes a Hypothesis Testable?

A good hypothesis must be testable: it must be possible, at least in principle, to collect data that would support or refute it. This seems obvious, but many statements turn out to be untestable upon closer examination.

4.1.1 Normative Statements

Consider the following:

“Crime should not be punished by death.”

“People should recycle to save our planet.”

These statements cannot be tested because they contain no observable implication. They express values, preferences, or moral judgments about what *ought* to be, not claims about what *is*. You cannot collect data to refute a “should” statement; you can only agree or disagree with the underlying values.

Normative commitments can certainly guide our research questions and our intellectual interests. Many researchers study capital punishment precisely because they hold strong moral views about it. But we must avoid systematically biasing our research design based on our normative preferences. Even if we feel strongly against the death penalty, our empirical and theoretical examination must be free of normative bias, and we should attempt to examine these issues as objectively as possible.

The distinction is crucial. “*The death penalty should be abolished*” is a normative claim. “*The death penalty does not deter murder more effectively than life imprisonment*” is an empirical claim that can be tested with data on crime rates and sentencing policies.

4.1.2 Statements Without a Baseline

“There are more people in the world today.”

The main issue here is that we have no comparison against which to draw a conclusion. More than *what*? More than yesterday? More than ever before? More than some specific past date? While it may not be a particularly interesting task, this statement can in principle be turned into a testable proposition by adding a baseline:

“There are more people in the world today than 100 years ago.”

Now we have a testable hypothesis. We can compare the current global population to documented population figures from a century ago and determine whether the statement is true or false. The addition of a specific comparison point makes the claim falsifiable — we could in principle discover that it is wrong, even though in this case we know it is correct.

4.1.3 Vague or Ambiguous Statements

“Economic conditions affect political stability.”

This is too vague to test. What economic conditions? GDP growth? Unemployment? Inequality? Inflation? What aspect of political stability? Regime survival? Protest frequency? Coup risk? Electoral volatility? Does “affect” mean increases, decreases, or merely relates somehow? The statement is theoretically plausible but empirically difficult to test because it does not specify what relationships to look for.

“Economic inequality increases the risk of political instability when economic growth is low.”

Now we have something testable. We can measure inequality (perhaps with Gini coefficients), identify periods of low economic growth (perhaps GDP growth below 2%), measure political instability (perhaps by counting protests, riots, or regime changes), and test whether the predicted relationship holds.

4.2 Characteristics of Good Hypotheses

i Four features of a well-formulated hypothesis

Specificity. It identifies concrete variables and the predicted relationship between them. Not “social factors matter for outcomes” but “social support reduces depression severity.”

Directionality. Where possible, it specifies not just *that* variables relate but *how*. Does *X* increase *Y*, decrease *Y*, or have no effect? “Higher education is associated with lower mortality rates” is better than “education and mortality are related.”

Falsifiability. It must be possible to imagine data showing the hypothesis is wrong. If no conceivable observation could refute it, it is not testable. “Either economic conditions affect stability or they don’t” is unfalsifiable — any observation confirms it.

Observability. The variables must be measurable, at least in principle. “Cosmic consciousness influences societal harmony” is not testable because we cannot observe or measure “cosmic consciousness” in any agreed-upon way.

4.3 Deriving Hypotheses from Arguments

The relationship between arguments and hypotheses is hierarchical and directional: **arguments come first and generate hypotheses**. You do not develop hypotheses and then look for arguments to justify them. You do not collect data, find patterns, and then generate post-hoc hypotheses to “predict” what you already observed. You start with theory, derive predictions from that theory, and then test those predictions.

Let us return to Humphreys and Weinstein. Recall that they articulated three distinct theoretical frameworks for why individuals join rebellions. Focusing on the grievance framework, they distilled the scholarship into three main hypotheses corresponding to the main explanatory strands they had articulated. Together, these theoretical variants all imply that an individual’s social position determines their propensity to participate in violence. From this foundation they derived:

Hypotheses

H1. Individuals are more likely to join a rebellion if they are economically deprived.

H2. Individuals are more likely to join a rebellion if they are marginalized from political decision-making.

H3. Individuals are more likely to join a rebellion if they are alienated from mainstream political processes.

Notice the tight correspondence between the argument and the hypotheses. The authors did not introduce hypotheses for which they had not previously articulated an argument. They did not suddenly test something they never discussed theoretically. Each hypothesis flows directly and obviously from one of the theoretical perspectives they developed in detail.

Also notice what the hypotheses *do not* include. They do not say “because they are desperate for resources”. Those causal mechanisms and scope conditions were discussed in the argument. The hypotheses are clean predictions that can be tested empirically.

This is the proper relationship between argument and hypothesis. The argument does the heavy lifting of explaining why relationships should exist. The hypotheses translate those theoretical expectations into testable predictions about what we should observe in data. Students often want to compress everything into the hypothesis, making it long and complicated. **Keep your hypotheses simple and your arguments nuanced.**

4.4 When Multiple Arguments Generate Competing Hypotheses

Different theoretical arguments predict may opposite relationships, and generate competing hypotheses. It sets up a theoretical test where data can help adjudicate between rival explanations.

Humphreys and Weinstein’s study exemplifies this. Their three frameworks do not all make the same predictions. The grievance framework predicts that the most deprived individuals join. But the selective incentives framework might predict that individuals with *moderate* opportunities are most likely to join — the desperately poor may be too risk-averse or incapacitated, while the well-off have good alternatives. The social sanctions framework might predict that social *integration* (the opposite of alienation) increases joining, because peer networks facilitate recruitment.

When you have competing theoretical frameworks, you can present competing hypotheses explicitly:

- **Grievance hypothesis:** Economically deprived individuals are more likely to join rebellions.
- **Opportunity cost hypothesis:** Individuals with moderate economic opportunities are more likely to join than either very poor or well-off individuals.
- **Network hypothesis:** Socially integrated individuals are more likely to join rebellions.

4.5 Common Issues in Hypothesis Formation

4.5.1 Issue 1: Hypotheses That Are Too Vague

“Social factors influence political outcomes.”

This tells us nothing testable. What social factors? Which political outcomes? What kind of influence? The hypothesis needs to specify variables and relationships clearly enough that we know what to measure and what pattern in the data would support or refute it.

4.5.2 Issue 2: Hypotheses That Are Too Complicated

“Economic deprivation increases rebellion participation, but only among politically marginalized individuals in contexts of ethnic tension when rebel groups offer selective incentives and the state is weak.”

This is actually several hypotheses bundled together, testing a complex interaction. While such complexity might be theoretically important, it should be built up from simpler hypotheses. Start with the main effects, then test moderators and interactions:

- **H1:** Economic deprivation increases rebellion participation.
- **H2:** The effect of economic deprivation is stronger among politically marginalized individuals.
- **H3:** The effect of economic deprivation is stronger in ethnically divided societies.

These can be tested sequentially, building from simple to complex.

4.5.3 Issue 3: Hypotheses That Include Mechanisms

“Economic deprivation increases rebellion participation *because it creates desperation*.”

The hypothesis should state what relationship you expect, not why you expect it. The mechanism (desperation) belongs in the argument. The hypothesis should read simply: “Economic deprivation increases rebellion participation.” You explained in your argument that the mechanism is desperation; the hypothesis just predicts the relationship you should observe if that mechanism operates as theorized.

4.5.4 Issue 4: Tautological Hypotheses

“People who participate in rebellions are more rebellious than those who don’t.”

A tautology is a statement that is true by definition — circular, and therefore untestable. This hypothesis essentially restates the dependent variable as both cause and effect. “Rebellious” and “participates in rebellions” are not distinct concepts; one defines the other. A genuine hypothesis must predict an outcome from factors that are conceptually and empirically separate from that outcome. Economic deprivation, political marginalization, and social networks are distinct from rebellion participation and can therefore genuinely predict it. But defining participants by a characteristic that simply describes participation adds no explanatory power. It is a logical circle that tells us nothing about why some people join rebellions while others do not.

4.5.5 Issue 5: Post-Hoc Hypotheses

Looking at your data, finding that age relates to rebellion participation, and then writing: *“Hypothesis: Younger individuals are more likely to join rebellions.”*

This is not hypothesis testing — it is describing patterns you already observed. True scientific hypothesis testing requires deriving predictions from theory *before* looking at data, then seeing whether the data match the predictions. Post-hoc hypotheses created to fit observed patterns do not constitute genuine tests.

4.6 How Many Hypotheses Are Necessary?

The number of hypotheses you need is not determined by arbitrary rules or conventions. It flows naturally from the structure and complexity of your theoretical argument. The question is not “how many hypotheses are enough?” but rather “how many distinct testable predictions does my argument generate?”

Too few hypotheses might signal that your theoretical framework is underdeveloped, or that you have not fully thought through the implications of your argument. A rich theoretical argument typically generates multiple predictions, and collapsing these into a single hypothesis can obscure important distinctions. If you find yourself with only one hypothesis, ask whether your argument is sufficiently developed or whether you are artificially bundling distinct predictions together.

Too many hypotheses creates the opposite problem: it suggests you are trying to test everything simultaneously without clear theoretical priorities. When hypothesis lists become unwieldy, you are often conflating different types of predictions — main effects that are central to your argument, interaction effects that specify boundary conditions, mediating mechanisms that explain how effects

work, and scope conditions that identify when relationships hold. Not all of these need separate hypothesis statements. Some are better discussed as theoretical nuances or empirical extensions than as core predictions requiring dedicated hypotheses.

The key is distinguishing between your central theoretical predictions — the primary relationships your argument claims exist — and refinements or complications of those predictions. Your main hypotheses should capture the core claims. Interactions, moderators, and mechanisms can often be discussed as theoretical elaborations without separate numbered hypotheses, or introduced as secondary hypotheses that refine your primary predictions.

Think about it this way: if someone asked you “what does your theory predict?”, what would you say? The answer to that question — probably articulated in a few sentences — should roughly correspond to your hypothesis set. If you need several minutes and many qualifications to explain all your hypotheses, you have likely over-specified.

The natural number varies by research context. A focused research paper testing one theoretical framework against alternatives might have a handful of competing hypotheses. A comprehensive study examining multiple dimensions of a complex phenomenon might have more, perhaps organized into families corresponding to different aspects of the theoretical argument. A dissertation or book-length project might include several hypothesis sets across different analytical chapters, each testing different components of an overarching framework.

But regardless of scope, the principle remains: **let your argument determine your hypotheses, not arbitrary numerical targets.** Each hypothesis should represent a distinct, meaningful theoretical claim that requires empirical evaluation. If you are generating hypotheses because you feel you “need more” rather than because your argument genuinely predicts multiple relationships, you are doing it wrong. Conversely, if your argument makes several distinct claims but you are forcing them into fewer hypotheses because you worry about having “too many,” you are obscuring important theoretical distinctions.

The test of the appropriate number is clarity and focus. Can a reader understand what your theory predicts by reading your hypotheses? Can they see the logical connection between argument and prediction? Do your hypotheses organize your empirical analysis in a meaningful way? If yes, you probably have the right number — whether that is two or ten. If your hypotheses seem arbitrary, redundant, or disconnected from your argument, the problem is not the number; it is the relationship

between theory and prediction. Quality trumps quantity.

4.7 Organizing Hypotheses in Your Writing

When presenting hypotheses in a research paper or proposal, clarity and organization matter.

Number your hypotheses clearly (H1, H2, H3, or Hypothesis 1, Hypothesis 2, Hypothesis 3).

This makes it easy to refer back to specific hypotheses when presenting results.

Present them after developing the argument they are derived from. Do not list all your hypotheses at the beginning of your paper and then explain why. First explain the theoretical logic, then state the hypothesis that flows from that logic.

Group related hypotheses together. If you have three hypotheses stemming from grievance-based arguments and three from selective incentive arguments, present them in those groups, not randomly mixed.

State hypotheses clearly and consistently. Use similar grammatical structure across hypotheses. Each should clearly identify the independent and dependent variables and the predicted relationship.

In your results section, return to each hypothesis explicitly. Do not just present your statistical analysis — organize your results around testing your hypotheses: “Hypothesis 1 predicted that economic deprivation would increase rebellion participation. Table 2 shows...” This creates clear correspondence between theory, prediction, and evidence.

5 Measurement: From Abstract Concepts to Observable Data

From your argument you have identified concepts — poverty, political marginalization, rebellion participation and need to figure out how to measure them in the real world.

Measurement is the process of translating abstract theoretical concepts into concrete, observable indicators that can be recorded as data. This translation is never perfect. Every measurement is ultimately a proxy for the underlying concept we actually care about, and every proxy has limitations.

Even something as straightforward as temperature as a theoretical concept is a case in point. What we care about is the average kinetic energy of the molecules in a substance, which nobody observes directly. A mercury thermometer instead exploits the fact that mercury expands when heated, and

we read the height of a column of liquid off a scale. The height is not the temperature; it is an indicator that stands in a known relationship to it. The proxy also has limits. Mercury freezes around -39°C and boils at 357°C , so outside that range the instrument tells you nothing, and thermocouples, infrared sensors, or resistance thermometers are needed instead. Each of these captures the same concept through a different physical mechanism, and each has its own range and error properties.

Social science measurement differs in degree rather than in kind. The relationship between our indicators and our concepts is less precisely known, and the concepts themselves are often contested in ways temperature is not.

Consider poverty. What does poverty really mean? Is it about absolute income? Relative income compared to others? Access to basic necessities? A subjective feeling of deprivation? Assets and wealth rather than income? Scholars have developed dozens of different poverty measures because the concept is multi-faceted and cannot be captured by a single measurement.

You can browse the World Bank's Poverty and Equity database to see how much measurement scrutiny theoretical concepts like poverty receive. You will find infant mortality rates used as poverty indicators, along with teenage pregnancy rates, household income levels, consumption expenditures, asset ownership, education levels, and many more. Researchers have agreed that poverty is too multi-faceted to be captured by a single measurement and that simple economic indicators are insufficient. Each measurement captures a different facet of the underlying concept, and researchers often use multiple measurements to triangulate on what they really care about.

Let us return to Humphreys and Weinstein to see how they translated their hypotheses into actual measurements. Since their question dealt with individual reasons for joining a rebellion, they chose to measure their concepts using survey questions administered to former combatants and non-combatants in Sierra Leone.

5.0.1 Measuring Economic Deprivation (H1)

For their first hypothesis, they needed to assess economic deprivation *as it was perceived before the war*. They used two separate measurements.

First, they coded whether individuals lived in a house with **mud walls** (1 if yes, 0 otherwise). This is a clever measure because in Sierra Leone housing material reflects wealth: wealthier families

have houses made of more durable, expensive materials. Mud walls indicate material poverty in a way that is relatively objective and not subject to people exaggerating or minimizing their pre-war economic status in retrospect.

Second, they used **education** as a proxy for wealth, coded as a three-category variable: more than primary education, primary education, or no education. In developing contexts, education strongly correlates with economic status because attending school requires that families can afford to keep children out of the workforce. Education also affects future earning potential, so it captures both current and future economic position.

Notice that they used two different measurements for the same underlying concept. Neither is perfect. Someone might live in a mud-wall house but have other assets. Someone might have education but still be poor. But together these measurements triangulate on material deprivation more effectively than either would alone.

5.0.2 Measuring Political Marginalization (H2)

For their second hypothesis, the goal was to measure the extent to which participants felt they were on the losing political side and excluded from decision-making. The authors drew on specific features of Sierra Leone's political history. The Sierra Leone People's Party (SLPP) had reportedly lost much of its influence to the ruling All People's Congress before the war, and many SLPP supporters belonged to the Mende ethnic group, thus strengthening their marginalization across ethnic lines.

Given this context, they used two measurements: whether someone **supported the SLPP** before the war (1 if yes, 0 if no), and whether someone belonged to the **Mende ethnic group** (1 if Mende, 0 otherwise). These measurements work specifically because of Sierra Leone's political history — they would not make sense in a different context. This shows how measurement must be tailored to the specific setting you are studying, drawing on local knowledge about what indicates the theoretical concepts you care about.

5.0.3 Measuring Social Alienation (H3)

The third hypothesis addressed general political alienation rather than the specific marginalization of particular ethnic groups or political supporters. Here the authors used a broader measure: whether individuals **supported any political party** before the war (1 if yes, 0 if no). The logic is that people who felt completely disconnected from the political system would not have bothered

supporting any party, while those integrated into political networks would have had party affiliations.

5.1 What Makes Measurements Useful?

All of these measurements were chosen to directly capture the concepts stated in the hypotheses, which themselves came from the theoretical arguments. There is a clear thread connecting theory to hypothesis to measurement. **Nothing came out of nowhere.**

The authors selected multiple measurements of the same concepts to more fully understand the relationships discussed in their argument. There is no firm rule about how many measurements you should choose per concept. These decisions must strike a delicate balance between accuracy and feasibility or cost. You need to weigh how much accuracy you gain by adding an extra measurement against the extra cost — in time, money, or respondent burden — that you must pay for that information. It is up to you as a researcher to make these judgment calls.

Also notice that the measurements are appropriate to the context. The authors did not use income in dollars: many rural Sierra Leoneans do not have regular cash income and might not be able to report it accurately. Instead they used housing materials and education, which people can reliably report and which meaningfully indicate economic status in that setting. Good measurement requires understanding the specific context you are studying.

Justify and criticize your own measures

Any discussion of your measurement must include critical justification of how well your concepts are captured by those measurements, and you should openly discuss the limitations. Mud walls indicate poverty, but they are not poverty itself — they are a proxy. Someone could be poor without living in a mud-wall house; someone in a mud-wall house could hold other forms of wealth. Education proxies for economic status, but imperfectly: scholarships exist, and education does not guarantee wealth. Being transparent about these limitations does not weaken your study. It demonstrates methodological sophistication and helps readers interpret your findings appropriately.

6 Data Collection: Gathering Your Materials

Once you have decided on your measurements, you are ready to collect your data. Notice again the sequence: research question first, then argument, then hypotheses, then measurement decisions, and only then data collection.

6.1 The Unit of Analysis Problem

It is crucial that you choose an appropriate **unit of analysis** for your data. The unit of analysis is the level at which your concepts are measured. In the Humphreys and Weinstein study, the unit of analysis was an individual. They surveyed individuals about their own experiences, attitudes, and behaviors. This makes sense because their theory was about individual-level decisions to join rebellions.

But not all research questions call for individual-level data. If you are studying why some countries develop faster than others, your unit of analysis may be the country. If you are studying how neighborhood characteristics affect crime rates, your unit of analysis might be the neighborhood. If you are studying marital satisfaction, you might need couple-level data.

Once you devise your measurement, the unit of analysis becomes much clearer. If your measurements are about individual characteristics, you need individual-level data. If your measurements are about organizational policies, you need organization-level data. **The measurement dictates the unit of analysis.**

6.2 A Cautionary Tale: The Division of Labor Study

Consider a team of researchers who started collecting data on the division of labor in families without having a clear idea what their argument was, or what their hypotheses and measurements should be. They were interested in how household labor is divided between mothers and fathers, but they had not thought carefully about what “division of labor” means or how to measure it.

In each household they visited, they collected information on the hours spent by only *one* household member on specific tasks. Sometimes they surveyed the mother, sometimes the father, but they never asked both parents in the same household. They thought this would be fine: they would have data on lots of mothers and lots of fathers.

Unfortunately, once they finished data collection, they realized they were simply unable to address

the issue of labor division, because they possessed information on either the mother or the father in each household, but never both. To understand *division* — how labor is split between two people — you need information about both people in the same household. You cannot compare mothers in Household A with fathers in Household B and learn anything meaningful about how couples negotiate household labor.

A thoughtful theoretical understanding of what division of labor means, developed prior to collecting data, would have ensured that they collected data per household in addition to individual-level information on both mothers and fathers. The concept “division” implies a relationship between two people, which requires measuring both members of the dyad. Had they worked through their argument and measurement first, this would have been obvious.

This is why rushing to data collection is dangerous. These researchers wasted time and resources collecting data that could not answer their question. They would either have to start over or abandon the project — and all of it was avoidable.

6.3 Defining Your Population

Your research question and argument are essential to determining what population you should sample from. This is another decision that flows directly from the theoretical work you have done.

The **theoretical population** is the set of cases your argument claims to be about — everyone the explanation should apply to.

The **target population** is the set of cases you can actually reach and enumerate. The first is defined by your theory; the second by practical access, existing records, and the limits of your resources. They are rarely identical, and the gap between them is where most overreaching in the conclusions section originates. Separating between theoretical and target population enables you to articulate the limits of your theory given the data you collected.

Suppose you are interested in what factors lead to the election of democratic presidents. You can stipulate your argument only with respect to current and past U.S. presidents, or you can attempt to answer the question for all presidents across all democratic countries. The population differs dramatically between these scenarios. In the first, it is known and finite: the United States over time and all presidents elected so far. You could potentially collect data on every case, making this a census rather than a sample. In the second, you must include every democratic country with a

presidential system, current and historical. The population is now much larger and more diverse, and you may need to sample, since collecting data on every president of every democracy would be prohibitively time-consuming.

Your theory may be about all presidential democracies, but records are patchy for early cases and for countries whose archives have not survived, and the cases you can actually enumerate will lean toward wealthier, more recent, better documented states.

6.4 Sampling from Your Population

Once you have defined the correct population, you can start drawing your sample. The fundamental idea is that you select the type of sampling method that provides the most representative cross-section of the population of interest.

In many instances, simple random sampling without any clustering or stratification does not accomplish this goal particularly well. By mere chance you might oversample some categories of your population — certain genders, geographic areas, or age groups — simply because of random variation. When you know certain characteristics are important (and your argument usually tells you this), you should correct for such issues by making sure you select from different groups proportionately. This is called **stratified sampling**, and it often produces more representative samples than pure random sampling, especially with smaller sample sizes.

In many cases, however, while we have a theoretical understanding of what our population looks like, we cannot know exactly what it is in practice. Consider a study of the effect of housing assistance on the wellbeing of HIV patients. The population of HIV patients is estimated and ultimately unknown. There is no master list of all HIV-positive individuals that is complete to sample from: many people do not know their status, others know but have not disclosed to any official registry, and privacy concerns prevent public health authorities from sharing patient lists with researchers.

In such cases, researchers often resort to **convenience sampling** or **snowball sampling**, asking participants to refer others with the same condition. These methods introduce biases — people in treatment networks are systematically different from those outside them, and people willing to be referred differ from those who are not — but sometimes they are the only feasible option. When using imperfect sampling methods, you must be explicit about the limitations they impose on your ability to generalize findings.

Key Terms

Research question. The focused question about the world that a study sets out to answer; it must balance ambition with feasibility.

Dependent variable. The outcome to be explained; the anchor for the argument, hypotheses, and measurement decisions that follow.

Argument. A theoretical explanation of why and how phenomena relate to one another, including the causal mechanism.

Mechanism. The specific process through which one phenomenon produces change in another.

Hypothesis. A specific, falsifiable prediction derived from an argument about what should be observed in data.

Tautology. A statement true by definition, and therefore untestable.

Post-hoc hypothesis. A prediction written after observing the pattern it “predicts”; not a genuine test.

Measurement (operationalization). The translation of an abstract concept into an observable indicator.

Proxy. An observable indicator that stands in for a concept it cannot perfectly capture.

Unit of analysis. The level at which concepts are measured — individual, household, neighborhood, organization, country.

Population. The full set of cases about which the argument makes claims.

Stratified sampling. Sampling proportionately from known subgroups to improve representativeness.

Convenience and snowball sampling. Non-probability methods used when a sampling frame is unavailable; they limit generalizability.